

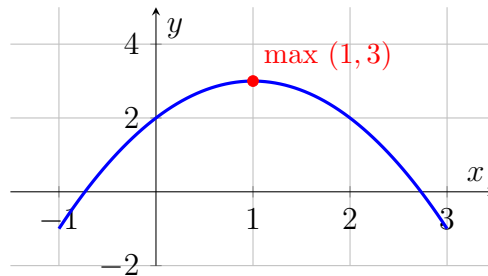
Final answers (record here — graded on the answer; show your work):

1. _____ 2. _____ 3. _____ 4. _____ 5. _____ 6. _____ 7. _____
8. _____ 9. _____

Increasing, Decreasing, Max & Min

Example (State max/min, vertex, and intervals):

For $y = x^2 - 4$, vertex form gives vertex $(0, -4)$. Leading coefficient $+1$ is positive, so it's a **minimum**. It decreases for $x < 0$ and increases for $x > 0$.



$y = -(x - 1)^2 + 3$ — opens down, so the vertex is a maximum.

Directions: State max or min, give the vertex, and give both intervals using inequality notation.

1. $y = x^2 + 5$

4. $y = -(x + 2)^2 - 1$

2. $y = -x^2 + 1$

5. $y = (x + 4)^2 - 6$

3. $y = (x - 3)^2$

6. $y = -(x - 1)^2 + 3$

7. $y = x^2 - 8x$

8. (challenge) $y = 2(x - 3)^2 + 1$

Challenge (same sheet):

9. (challenge) For $y = -2(x + 3)^2 + 7$, find the vertex, state max or min, give both intervals, AND explain what would change about your answer if the leading coefficient became positive instead.

Answer Key — fully worked

Procedure: Write the equation in vertex form $a(x - h)^2 + k$ if it isn't already.; The vertex is (h, k) .; If $a > 0$ the parabola opens up (minimum); if $a < 0$ it opens down (maximum).; The graph decreases before the vertex and increases after it (for a minimum) — reversed for a maximum..

1. Opens up, minimum; vertex $(0, 5)$; decreasing $x < 0$, increasing $x > 0$
2. Opens down, maximum; vertex $(0, 1)$; increasing $x < 0$, decreasing $x > 0$
3. Opens up, minimum; vertex $(3, 0)$; decreasing $x < 3$, increasing $x > 3$
4. Opens down, maximum; vertex $(-2, -1)$; increasing $x < -2$, decreasing $x > -2$
5. Opens up, minimum; vertex $(-4, -6)$; decreasing $x < -4$, increasing $x > -4$
6. Opens down, maximum; vertex $(1, 3)$; increasing $x < 1$, decreasing $x > 1$
7. Complete the square: $x^2 - 8x = (x - 4)^2 - 16$; minimum; vertex $(4, -16)$; decreasing $x < 4$, increasing $x > 4$
8. $a = 2 > 0$ opens up, minimum; vertex $(3, 1)$; decreasing $x < 3$, increasing $x > 3$
9. Vertex form $a(x - h)^2 + k$ gives vertex $(-3, 7)$. Since $a = -2 < 0$ the parabola opens down, so $(-3, 7)$ is a maximum. It increases on $x < -3$ and decreases on $x > -3$. If the leading coefficient became positive (e.g. $+2$), the parabola would open up instead: the vertex $(-3, 7)$ would become a minimum, and the intervals would flip — decreasing on $x < -3$ and increasing on $x > -3$.